

The results — a completed four-setting study

The empirical realisation of the [research programme](#) in [ad hoc semantic reconciliation](#). Four operational settings, built, run against a validated gold standard, and reported, with a master report drawing them into one synthesis.

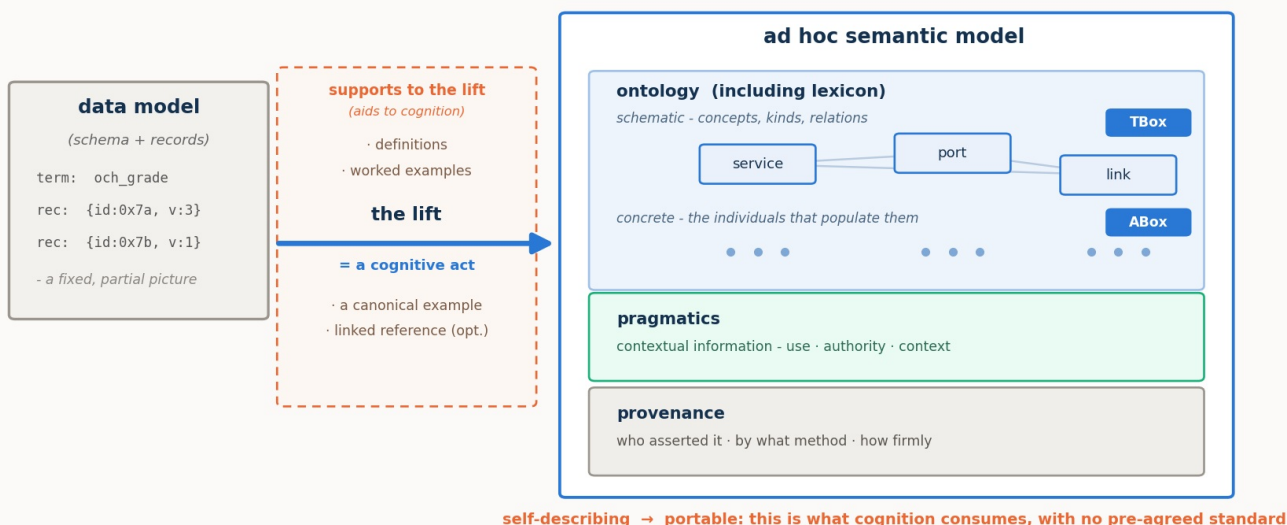
Part of the larger project — the theory and programme that motivate this study are in [../proposal/](#) ; the interactive demonstrations are in [../demos/](#) .

The central finding, established across all four settings: **it is cognition that completes a reconciliation**. Descriptor methods — matching names, then names plus a gloss — carry it to a ceiling and stop, historically leaving the rest to a standard or a person. Where both systems can reason, the remainder is closed **autonomously**, with no model agreed in advance and no human in the loop, and only as cognition recedes does a residual have to be referred onward. A thin, published **reference** can substitute for cognition, or supply an inert side the facts it lacks — but its reach ends there: it hands over information, never the **authority** to decide, whose value governs or whether a trade-off is acceptable. And the **pragmatic** layer — what a reconciled thing is *for*, whether it matters, who decides — is the frontier the descriptor methods never reach: decisive for meaning, and itself bounded by the agent's capability.

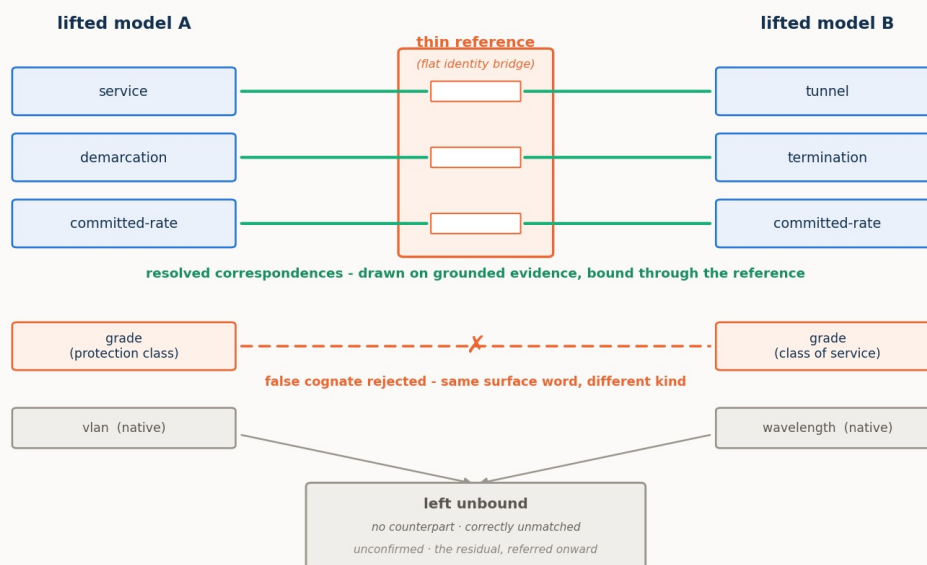
Start here — the master report

[reports/MASTER_REPORT.md](#) · [PDF](#) · [slides \(PDF\)](#) — the synthesis that sits atop the four settings. It introduces the idea once (the lift, portable semantic models, the family of reconciliation operations, and the cognition spectrum), distils each setting to its essentials, and then gathers the findings so they can be read as one result: what works and how far, the **six theses** the settings establish, the evidence behind them along four cross-cutting axes (cognitive load, the reference, placement and failure, and the pragmatic layer), two tables mapping every finding to its placement and to the process stage it acts on, the surprises, and the scope. The slide deck carries the same synthesis for readers who prefer slides.

The lift - cognition turns a data model into an ad hoc semantic model (ontology, pragmatics, provenance)



Reconciliation over two lifted models - grounded correspondences bound through a thin reference, a rejected cognate, and what is honestly left unbound



The four settings

Each setting takes the same frame to a new operation. Each has a report (renders inline), a print-ready **PDF**, and an interactive **demo** (opens in your browser; see [./demos/](#) for download links too).

1/4 • Configuration • PDF • ► demo — two standard models of one network. ONF **TAPI** ↔ IETF **TEAS/ACTN**, seeded with false cognates and opaque items. The founding setting: it shows the thin reference **substituting for cognition**, the benefit **capability-dependent**, verification catching the traps, and work scaling **linearly** with a reference against quadratically without. Its four sub-studies are folded into the report and preserved as method notes under [notes/studies/](#).

2/4 • Intent • PDF • ► demo — refinement, negotiation, and a service that renegotiates itself. A customer's intent reconciled against an operator's catalogue by **refinement**, not equivalence; verification becomes a **satisfaction** check; a two-sided **negotiation** appears; the pragmatic operation enters as a portable **movable policy**; and the exchange **recurs across a service's life**. Grounded in the IRTF NMRG draft *draft-janz-nmrg-naas-agentic-negotiation*.

3/4 • Cross-domain • PDF • ► demo — reconciling with no public standard. Two **home-grown, private** models meet at one seam. The central result is a **mirror**: without the constructed reference the strong agent **under-commits** (defers at perfect precision) and the weak agent **mis-commits** (binds wrongly). A single descriptive field unlocks a capable agent; a bare shared pointer is worse than nothing. Building the shared ground is the work — and, run end to end, an agent that builds it itself lifts from **0.20 to 0.80** (0.90 with a decisive virtual experiment).

4/4 • Observability • PDF • ► demo — an alarm is not an anomaly. A legacy fault manager and an IETF **NMOP** agent (RFC 9940) reconcile two observability worlds, carrying the programme's deepest false cognate and a one-to-many decomposition. The **pragmatics carry the operative verdict** — but only for an agent able to carry them; while **correlation**, a structural pragmatic, is robust across the whole ladder.

The reconciling agents are one **model ladder** for the whole study: **gpt-5.6-sol (strong)**, **gpt-5-mini (mid)**, **gpt-5-nano (weak)** — "sol / mini / nano" — chosen so the ladder isolates model strength rather than provider or architecture.

How it works

The **lift** is the move from a *data model* (a schema and its records: already a partial semantic picture, but a fixed one, with much of its meaning left implicit) to a *semantic model* that carries its meaning explicitly. A semantic model has three parts: an **ontology**, including its **lexicon**, with a schematic layer (concepts, their kinds and relations, and the labels and synonyms that name them: the **TBox**) and a concrete layer (the instances that populate them: the **ABox**); **pragmatics**, the contextual information a consumer needs (what a thing is for, whose authority governs it, in what context it holds), obtained not by lookup but by judgement; and **provenance** (who asserted each part, by what method, how firmly). The lift itself is a cognitive act, aided by supports that are not parts of the model: definitions, worked examples, a canonical example, and an optional link to a shared reference. Being self-describing, a lifted model is **portable**: any cognitive consumer can pick it up with no pre-agreed standard. Reconciliation runs over these lifted models, not over labels.

Drawn out, the ontology and its instances are a **knowledge graph**: the schematic layer is its schema (typed concepts and permitted relations) and the concrete layer is its assertions (typed individuals and the edges among them). Reconciliation divides the same way — aligning concepts is ontology matching, and co-referencing individuals is entity resolution over the graph — which is where the community's knowledge-graph work for network operations meets this approach. The pragmatics and provenance are what the semantic model adds beyond a bare graph, and they are exactly what a thin reference cannot supply.

A **case** is two lifted semantic models plus a gold standard *derived from the models and validated*, so it cannot drift. A **reasoning stack** reconciles them — deterministic controls and a language-model agent run at each point on the **cognition spectrum** (both live, one inert, both inert). At the fully-cognitive end the two live sides reconcile as two separate agents under information asymmetry, each holding only its own lifted model and a surface catalogue of the other's, proposing correspondences and ratifying the other's, with the option of a decisive virtual experiment to settle a hard pair; as a side goes inert a single agent reconstructs it. The **harness** scores the output against the gold and records quality — *precision*, *resolved fraction*, *surviving false cognates*, and the *residual* — and, for the agent, cognitive effort (reasoning tokens, latency). A resolved fraction below one is deferral, not error: the residual is the shortfall from full cognition's reach, and it grows as cognition recedes.

What's new (2026-09-10)

This revision brings the **pragmatic layer** to the front, shows the **lift itself** as an act of cognition, and makes the agent-performed lift **replicable from the command line**. The central findings are unchanged; the reports carry the results in place.

- **Pragmatics across the lifecycle — new master-report §13.9.** The report now separates the three layers of a lifted model by *how each is obtained from it*. The ontology's schematic layer (the **TBox**) and its instances (the **ABox**) are both *observed in the lifted model*: a determinate lookup any consumer, even a non-cognitive one, can do. Each is the frozen output of the cognitive lift that made it explicit, not a picture the source data model hands over ready-made; the cognition is front-loaded, not absent. The **pragmatics** (whether a state matters, whether an offer is acceptable, who decides) are *judged*, from schema, instances and context, and are defeasible and context-relative. The asymmetry is the point: TBox and ABox need cognition once, at lift time, and then freeze cleanly and read as lookup forever; pragmatics cannot be frozen cleanly and must be re-judged against the live context at each use. A static model, ours or an OWL export, holds the TBox in full and the ABox as a refreshable snapshot, but can only *describe* pragmatics, never carry them resolved; the section shows the OWL mapping and its limit. A live operation supplies them through four general mechanisms: pulling a pragmatic fact *in* by interrogating a live peer, *refreshing* it against the current context, *creating certainty* by a decisive virtual experiment, and *resolving* the verdict itself

by judgement, the last being the capability-gated frontier. The static artefacts a system carries turn out to be *frozen outputs* of these mechanisms (a published reference is a frozen pull-in, a snapshot a frozen refresh, a pre-placed policy a frozen resolve), each valid only until the context it assumed drifts; and the mechanisms are gated by two things at once, liveness (which recedes across the cognition spectrum, so the mechanisms fall away in order as sides go inert) and capability (which bites hardest on resolution). This is why the framework does not merely extend beyond reconciliation but predicts the other settings' signatures, the observability verdict being capability-gated while its correlation, a structural test, holds across the ladder. The point is carried through the rest of the report: §14's second map gains a note tying its **Lift** and **Pragmatic-resolution** rows to §13.8 and §13.9, and the one-paragraph close (§17) now states it.

- **The lift, shown and performed — extended §13.8.** Three things land in the lift section. A *concrete lifted model* is now printed in full (§2 shows one complete concept as JSON, with a note that the shape is encoding-agnostic), so "what exactly is in a lifted model?" has a definite answer. A *lift-in-the-act* trace box shows the agent naming its own per-concept evidence on the relabelled control, where the recognisable names are replaced by invented ones so only the structure remains: the agent names the surface signals it used and the look-alikes it ruled out, confirming the reading is structural, not recalled. And the agent-performed lift is run *lift-only across eleven schema cases and the whole model ladder*: coverage is complete and structure intact at every rung, every false-cognate trap is caught (the relabelled control included), and the differences are of verbosity, not accuracy — so §16 gains a clause that lift correctness is robust across the ladder.
- **New code, and the replicable work it supports.** The agent-performed lift is now a first-class, reproducible entry point. `pipeline/lift_study.py` gains `--lift-only` (produce the lift without running reconciliation), `--cases all` (discover and run every schema case), and `--trace` (record the agent's stated per-concept evidence); `src/reconcile/lift.py` gains the trace mode, and `src/reconcile/model.py` a `to_dict()` that serialises a lifted model back to the case JSON shape, with offline tests added under `tests/`. Running `pipeline/lift_study.py --lift-only --cases all` over the three-model ladder is what produced the eleven-case result above; the lifted models it emits are committed under `results/agent_lifted_models/` so the finding can be inspected and rebuilt.
- **Figures and deck updated to match.** Figure 1 now labels the ontology's schematic and concrete layers as the **TBox** and the **ABox**, tying the opening picture to §13.9. The constructed scaling chart (former Figure 11) is dropped — it was not a peer to the measured figures and its point is made in the text — and the later figures are renumbered. The master slide deck carries the new material: the pragmatics-across-the-lifecycle distinction, the four operation-time mechanisms, and the lift in the act.

What's new (2026-09-08)

This revision deepens the fully-cognitive end and adds two controls. As before, the central findings are unchanged and the reports carry the results in place; this note is for readers tracking what moved.

- **Completion by decisive experiment, shown end to end.** At the fully-cognitive end, when the two agents cannot settle a correspondence by discussion, either can provision the candidate in a private virtual copy of the network, operate it, and read back whether it holds; the verdict binds or refutes. With this, two agents reach the reference's close on three of the four schema settings with no reference at all, and the master report now includes verbatim transcripts of the negotiation and of the experiment. (See §13.7.)
- **The lift, performed by an agent.** The study reconciles lifted models whose explanation layer, each concept's gloss and worked example, had been supplied ready-made. An agent is now asked to produce that layer itself from a side's bare schema, and reconciliation over the agent's own lift matches the ready-made one: identical for the capable agent binding through a reference, across all four settings. So the results do not depend on the lift being authored in advance; what stays open is the lift from raw, messy schema and a cold start before any instances exist. (See §13.8.)
- **A memorisation control.** The flagship TAPI/TEAS case is re-run with the public-standard identities

stripped and relabelled, to separate reasoning from recall of a known standard: reasoning alone reaches a resolved fraction of about 0.67 where the recognisable case reaches 1.00, and a shared reference restores 1.00 either way. (See §16.)

- **Slide deck refreshed** to carry the negotiation, the decisive experiment, and the lift study.

What's new (2026-09-05)

This revision strengthens the study in-house — the central findings are unchanged and hold — and closes the in-house gaps the first cut left open, leaving only the genuinely external one (real network data). The reports carry the results in place; this summary is here for readers tracking what moved between revisions.

- **Construct-then-bind, end to end (setting 3).** The standard-free setting previously handed the agents the shared reference and scored the binding alone. The agents now **build that reference themselves** from the two models and then bind through it, with nothing pre-given. The strong agent lifts from a no-reference resolved fraction of **0.20 to 0.80** (0.90 with a decisive experiment), near the reference-given 1.00, at perfect precision and with no false cognate. Constructing the shared ground works, and building it is the work.
- **A six-model capability sweep.** Swept across six models at both-cognitive, two-agent negotiation **compresses** the capability gradient: bilateral ratification holds precision high and drives false cognates to zero across almost the whole ladder, while resolution is uniformly lower. The discipline a lone weak agent would lack is supplied by the negotiation structure itself.
- **Instance co-reference measured beyond setting 1.** Entity-level co-reference is now measured in settings 2 and 4 as well, reproducing the budget-limited-then-structural pattern: capable agents resolve fully where a live side can be interrogated, the weakest only partially. (A fix to the interrogation path — letting an agent ask which attributes exist rather than guess field names — moved setting 4 from a floor to full resolution.)
- **A classical descriptor baseline.** A non-cognitive matcher (labels, glosses, structure) is included as an explicit baseline, confirming that descriptor methods carry a reconciliation to a ceiling that cognition then passes.
- **A reach study — what an agent can see and what it can ask.** Holding the strong agent at its best, we vary two things independently: how much of a record it may *see* (from a bare key up to full topology) and what it may *ask* of the live system (nothing; a named attribute; the open question "what facts do you have?"; a full virtual experiment). Seeing more, on its own, caps the hardest cases at one-half. The hinge is the open question: being able to ask what exists — rather than having to name the field in advance — takes the reconciliation to a full, correct close. Run down the model ladder, the same access lands by capability: the strong agent converts it to a full close, the mid plateaus, and the weakest posts high resolved fraction but with slipping precision. (See §13.5 of the master report, Figure 14.)
- **Breadth — every signature reproduced on new cases.** Each of the four settings is now exercised on **three independently-built cases, not one** — two new ones per setting, in different domains, with different vocabularies and different traps. Every signature reappears: cognition completes the match while the reference mainly prevents the weaker agents' errors (configuration); the strong-omit / weak-commit mirror, repaired by a thin reference (cross-domain); refinement and negotiation completing while the multi-hop lifecycle grades with capability (intent); and the alarm-is-not-an-anomaly look-alike reliably told apart (observability). This closes the "you only showed it on one case" worry as far as in-house work can. (See §16 of the master report, Figure 16.)
- **The economics of the reference — when constructing it is worth the cognition.** Building the shared reference is itself an act of cognition, so we measured its cost against what it buys. The rule: it is load-bearing only where **no standard exists and the agent is capable** (cross-domain, strong agent: resolved fraction 0.20 → 0.80, and 0.90 with a decisive experiment, for a little extra spend); where a reference already exists (configuration, observability) the **given** reference reaches the

same close for a fraction of the cognition, so constructing one is **redundant**; and for a **weak agent it is counterproductive** — construction cost explodes (to tens of thousands of reasoning tokens) for a worse close, not a better one. So construct-then- bind is not a blanket default: build the reference where there is none to bind through and the agent can build a good one; otherwise use the one that exists. (See §13.6 of the master report, Figure 15.)

- **Benchmark packaged.** The cases, harness, and leaderboard are packaged under `benchmark/` for external use — now eighteen cases across the four settings.

What remains is now genuinely external: larger and more varied cases drawn from real networks, and the real-data grounding that only carrier and standards-body collaboration can supply. The breadth cases above are still authored by the same hand, so they test robustness to *variation*, not to *real* data — which is exactly the gap that external collaboration exists to close.

Repository layout (`study/`)

reports/	the deliverables
MASTER_REPORT.md	the synthesis – start here
REPORT_1of4_configuration.md	setting 1 · configuration (TAPI ↔ TEAS)
REPORT_2of4_intent.md	setting 2 · intent (refinement, negotiation, lifecycle)
REPORT_3of4_cross_domain.md	setting 3 · cross-domain (standard-free)
REPORT_4of4_observability.md	setting 4 · observability (alarm ≠ anomaly)
notes/	
studies/	setting-1 method notes (lift baseline, reference anatomy, instance disambiguation, verification modes)
design/	design notes of record, per setting
archive/	superseded working notes
SETUP_OPENAI.md	how to supply the API key (kept out of the repo)
figures/	all report figures (regenerated by <code>pipeline/figures*.py</code>)
src/reconcile/	the harness: models, reference, metrics, stacks, oracles
benchmark/	case builders and cases/ (two lifted models, reference, traps, gold)
results/	the exact per-run data behind every figure and table
pipeline/	all runner, figure, and build scripts
tests/	offline tests of the stacks and scoring (no API)
pyproject.toml	

Quick start

Run from the `study/` directory. The controls, gold derivations, offline tests, figures, and PDF build need no model access (standard-library Python plus `matplotlib/markdown/wkhtmltopdf`). The language-model runs need `openai` and an API key — see [notes/SETUP_OPENAI.md](#) — and reach `api.openai.com`, so they run where that is available.

```

cd study

# deterministic, no API:
python pipeline/run.py --case config_big_hard --no-write      # controls only
python pipeline/scaling.py --max-n 12                        # the scaling result
python -m pytest tests/                                       # offline tests
python pipeline/figures_master.py                             # regenerate the master figures
python pipeline/build_pdfs.py                                  # rebuild every report PDF

# the language-model studies (need OpenAI; run where api.openai.com is reachable):
python pipeline/run.py --case config_big_hard --agent --trials 4 \
  --placement both_cognitive,one_inert,both_inert \
  --model gpt-5.6-sol,gpt-5-mini,gpt-5-nano
python pipeline/intent_study.py      --model gpt-5.6-sol,gpt-5-mini,gpt-5-nano
python pipeline/pragmatics_study.py  --model gpt-5.6-sol,gpt-5-mini,gpt-5-nano
python pipeline/observability_study.py --model gpt-5.6-sol,gpt-5-mini,gpt-5-nano

# the agent-performed lift, on its own, across every schema case (dumps each lifted model to JSON;
# add --trace to record the agent's stated evidence per concept):
python pipeline/lift_study.py --lift-only --cases all --model gpt-5.6-sol,gpt-5-mini,gpt-5-nano

```

Per-setting run commands are in each report's Reproducibility section, and the method notes under `notes/studies/` carry the sub-study commands.

Grounding and scope

The study operationalizes and empirically tests the claims of the [proposal](#): the IRTF NMRG work on agentic network-as-a-service negotiation, the cross-domain provisioning demonstration, and the IETF NMOP anomaly/alarm model (RFC 9940). The claims are **existential and mechanistic** — *this is how ad hoc reconciliation works, and here it is working* — established on single, seeded cases built to exercise each mechanism and prove each trap, across the model ladder; they are not population estimates. What remains is now external: larger and more varied cases drawn from real networks, and the real-data grounding that only carrier and standards-body collaboration can supply.

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